📊 Global E-Commerce Sales Analysis Project

1. Project Overview

This project focuses on analyzing global e-commerce sales data to uncover trends in sales performance, profitability, customer behavior, and regional distribution.

The analysis was conducted using:

- Power BI (for visualization)
- Power Query (for data cleaning and transformation)

2. Objectives

The main goals of this project were to:

- Analyze total sales, profit, and order volume
- Identify top-performing products
- Understand regional sales distribution
- Evaluate customer payment preferences
- Track sales trends over time

3. Data Cleaning & Preparation (Power Query)

Data cleaning was performed to ensure accuracy, consistency, and usability for analysis.

Steps Taken:

- Removed duplicate records
- Handled missing/null values
- Standardized column names (e.g., Total_Sales, Order_ID)
- Converted data types:
- Dates → Date format
- Sales/Profit → Numeric
- Created new calculated columns where necessary
- Ensured consistency in:
- Product categories
- Regions
- Payment methods

This step improved data quality and ensured reliable insights.

4. Data Modeling

Established relationships between tables (if applicable)

Ensured proper field categorization:

- Dimensions (Region, Product, Category, Date)

5. Dashboard Development (Power BI)

The dashboard was designed to provide clear, interactive insights.

🔹 Key KPIs

- Total Sales: 206.32K
- Total Profit: 66.72K
- Total Orders: 907

These KPIs give a quick overview of overall business performance.

6. Visualizations & Insights

📌 Sales by Region

Europe and North America generate the highest sales

Middle East & Africa show lower performance

Insight: Focus expansion strategies in high-performing regions.

📌 Top 5 Products by Sales

Wireless Bluetooth products lead significantly

Ergonomic and webcam products follow

Insight: Tech accessories dominate sales → potential for scaling

📌 Sales by Payment Method

Credit Card is the most used

PayPal and Cash on Delivery follow

Bank Transfer is least used

Insight: Optimize checkout experience for dominant payment methods

📌 Sales Trend Over Time

Sales fluctuate over months

Peak observed around mid-year periods

Insight: Possible seasonality in sales patterns

📌 Sales by Location (Map)

Sales are globally distributed

Higher density in developed regions

Insight: Market penetration varies geographically

7. Key Insights

- Strong performance in Europe & North America
- Wireless products are top revenue drivers
- Credit cards dominate payment preferences
- Sales show seasonal fluctuations
- Opportunity exists in underperforming regions

8. Evidence of Thinking

Defined the problem: How can we understand global sales performance?

Considered:

- Regional differences
- Product demand
- Customer behavior

Trade-offs:

- Focused on key KPIs for clarity instead of overcrowding dashboard

9. Evidence of Execution

- Cleaned raw dataset using Power Query
- Built calculated metrics:
- Total Sales
- Total Profit

- Total Orders

Designed interactive dashboard with:

- Filters (Year, Category, Region)
- Charts and KPI cards

10. Evidence of Impact

- Since this is a portfolio project, impact is demonstrated using insights:
- Identified high-performing regions → supports expansion strategy
- Highlighted best-selling products → informs inventory decisions
- Revealed payment trends → improves customer experience

11. Tools Used

- Power BI
- Power Query
- Excel/CSV dataset

12. Conclusion

This project demonstrates the ability to:

- Clean and transform real-world data
- Build interactive dashboards
- Extract actionable business insight

13. Future Improvements

- Add customer segmentation analysis
- Include predictive analytics (forecasting sales)
- Integrate real-time data sources